ARK-489 — Authority Engine · Burst Throughput

Status: PREREGISTRATION (locked before execution)
Experiment Date: 2026-07-18
Series: ExecutionProof P02 Latency/Throughput/Scale

Question

What is the peak (burst) throughput — authority decisions/second in a short high-intensity interval — of the reference **Authority Engine**?

Analogous to ARK-484 (Verification Decision burst), applied to the current-state authority check.

Scope & Honesty Boundary

Minimal in-process reference `AuthorityEngine`, built to MEASURE, not the production engine. Claims bounded to this reference under the stated load.

Hypothesis

Burst throughput **≥ 200,000 decisions/second** over a 10-second window, at **100% accuracy**.

Component Under Test

Reference `AuthorityEngine` (exact, fail-closed). `engine/perf_harness.py --dimension burst`.

Methodology

- **Warmup:** 10,000 decisions (excluded)
- **Burst window:** 10 seconds of continuous decisions on the valid grant path
- Count total decisions and correct decisions; throughput = total / elapsed

Metrics

Primary: `burst_throughput_dps` (decisions/second), `accuracy`
Secondary: `total_decisions`, `duration_sec`

Thresholds (Gate Conditions)

C1 (Throughput): ≥ 200,000 dec/s
C2 (Accuracy): 100% (all decisions correct)
C3 (Correctness gate): valid→ALLOW, mutated→DENY, revoked→DENY

Verdict: PASS if C1 ∧ C2 ∧ C3; otherwise FAIL

Kill-Gate

K1: Accuracy < 100% → guard logic compromised → GATE-STOP.

Honest Findings Commitment

Any deviation disclosed in RESULTS.md.

Execution Protocol

1. **Lock** prereg + MANIFEST.txt hashes
2. **Execute** `engine/perf_harness.py --dimension burst`
3. **Record** to `results/burst_results.json`
4. **Evaluate** against C1–C3
5. **Report** in RESULTS.md

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Investigator: Remnant Fieldworks Inc. / ExecutionProof Research Program
Covenant: RF Standing Covenant (preserve outcomes, bound claims, honest disclosure)